Prabhavit Gupta

GenAI & Agentic Systems Engineer: Model Context Protocol (MCP), LangGraph & RAG Pipelines

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Summary

Hands-on GenAI Engineer with 3 years of commercial experience designing and delivering production-grade LLM applications, autonomous agentic workflows, and high-precision RAG architectures. Proficient in Python 3, FastAPI, Anthropic Claude, OpenAI GPT, and developing custom Model Context Protocol (MCP) servers using FastMCP to bridge AI agents with enterprise data and tool ecosystems. Proven expertise in building cyclical agent graphs with LangGraph, optimizing hybrid vector search (pgvector, Chroma), and implementing end-to-end observability, tracing, and evaluation frameworks (LangSmith, Langfuse, OpenTelemetry). Based in Venice, Italy with permanent EU work authorization (Permesso di Soggiorno UE a Tempo Indeterminato, Italian Partita IVA ready, immediate start).

Experience

FlowSpace.co • *Lead GenAI Engineer & Systems Architect (Freelance)*
Mar 2024 - Aug 2026 • London, UK (remote)

- Architected and deployed production LLM applications, conversational systems, and multi-agent workflows supporting 4,500+ active enterprise users with 99.9% uptime.
- Developed and maintained custom MCP (Model Context Protocol) servers using FastMCP and Python, exposing enterprise internal tools, schemas, and PostgreSQL databases to Claude and GPT models.
- Engineered high-precision RAG pipelines featuring document ingestion, hybrid semantic search, metadata filtering, and reranking to eliminate factual hallucinations.
- Built autonomous multi-agent reasoning workflows using LangGraph and Python, orchestrating dynamic tool execution, function calling, and structured JSON output validation.
- Built high-performance, asynchronous REST APIs using FastAPI, containerized with Docker and deployed across AWS and GCP environments.
- Implemented comprehensive LLM observability harnesses (LangSmith, Langfuse), tracking latency, token usage, evaluation benchmarks, and cost efficiency.
- *Tools: Python 3, FastAPI, FastMCP, Model Context Protocol, LangGraph, LangChain, RAG, pgvector, Claude, GPT, Docker, Git*

Urlink • *AI Solutions Developer & Integrations Engineer (Freelance)*
Apr 2025 - Jul 2025 • Barcelona, Spain (remote)

- Developed automated document extraction and classification pipelines connecting external partner APIs with centralized database stores.
- Implemented semantic chunking and structured prompt validation layers in Python, ensuring data integrity across high-volume ingestion streams.
- Partnered with product teams and client stakeholders to translate functional requirements into scalable AI architecture specifications.
- *Tools: Python, FastAPI, REST APIs, PostgreSQL, Supabase, Git, Linux*

Independent Technical Consultant • *GenAI & Automation Solutions Developer*
Sep 2023 - Mar 2024 • Venice, Italy

- Delivered custom conversational agents, AI-powered document search tools, and workflow automation microservices for European SMB clients.
- Implemented function calling, webhook listeners, and secure API integrations to connect LLMs with existing enterprise systems.
- Authored technical architecture runbooks, prompt engineering guidelines, and deployment specifications for client engineering teams.
- *Tools: Python, LangChain, OpenAI API, REST APIs, Linux, Git*

Tractionboard.co • *Low-Code Developer (Internship)*
Aug 2023 - Oct 2023 • Utrecht, Netherlands (remote)

- Built automated webhook consumers and data integration workers in Python for SaaS client onboarding flows.
- *Tools: Python, REST APIs, Sendgrid*

Education

Ca' Foscari University of Venice • *Bachelor's in Philosophy, International Studies & Economics, Minor in Entrepreneurship*
Oct 2020 - Jun 2024 • Venice, Italy

Certifications

Microsoft: Introduction to building AI agents • Microsoft Copilot Studio Agent Academy • Microsoft Power Automate • Tray.AI Build Practitioner (Sep 2024) • Tray.AI Build Foundations

Additional Information

- **Work authorization:** Permesso di Soggiorno UE per Soggiornanti di Lungo Periodo (Italy), unrestricted work authorization in Italy and across the EU (CCNL or Partita IVA). No visa sponsorship required.
- **Languages:** English (fluent / C1 professional), Hindi (native), Italian (working beginner).
- **Availability:** Available immediately for full-time remote roles in Europe.

Core Skills

- **Model Context Protocol & Tools:** Model Context Protocol (MCP), FastMCP, Skills / Plugins, Tool Calling, Function Calling, Structured JSON Outputs.
- **LLMs & Agent Orchestration:** Anthropic Claude, OpenAI GPT, LangGraph, LangChain, LlamaIndex, Multi-Agent Workflows, Cyclical StateGraphs.
- **RAG & Vector Architecture:** Retrieval-Augmented Generation, Document Ingestion, Semantic Chunking, Hybrid Search, Vector DBs (pgvector, Chroma, Qdrant).
- **Backend & Cloud Systems:** Python 3, FastAPI, Asynchronous Programming, REST APIs, PostgreSQL, Docker, AWS, GCP, Linux, Git.
- **Observability & Quality:** LangSmith, Langfuse, OpenTelemetry, Token/Cost Tracking, Regression Testing, Responsible AI Guardrails.